Homework Section 14.5

1. Use the Chain Rule to find $\frac{dz}{dt}$.

   a) $z = x^2 y + xy^2$, $x = 2 + t^4$, $y = 1 - t^3$

   b) $z = x\ln(x + 2y)$, $x = \sin t$, $y = \cos t$

2. Use the Chain Rule to find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$

   a) $z = \frac{x}{y}$, $x = se^t$, $y = 1 + se^{-t}$

   b) $z = e^r \cos\theta$, $r = st$, $\theta = \sqrt{s^2 + t^2}$

3. Use the Chain Rule to find $\frac{\partial z}{\partial v}$ for $z = \cos x \sin y$, $x = u + v^2$, and $y = u^2 - v^2$. Write your answer as a function of *u* and *v*.

4. Let $z = x^2 + xy^3$, $x = uv^2 + w^3$, and $y = u + ve^w$. Use the Chain Rule to find $\frac{\partial z}{\partial u}$, $\frac{\partial z}{\partial v}$, and $\frac{\partial z}{\partial w}$ when $u = 2$, $v = 1$, $w = 0$.

5. Let $R = \ln(u^2 + v^2 + w^2)$, $u = x + 2y$, $v = 2x - y$, and $w = 2xy$. Use the Chain Rule to find $\frac{\partial R}{\partial x}$ and $\frac{\partial R}{\partial y}$ when $x = y = 1$.

6. The radius of a right circular cone is increasing at a rate of 1.8 in/s while its height is decreasing at a rate of 2.5 in/s. At what rate is the volume of the cone changing when the radius is 120 in. and the height is 140 in.?

7. Wheat production in a given year, *W*, depends on the average temperature *T* and the annual rainfall *R*. Scientists estimate that the average temperature is rising at a rate of 0.15 degrees Celsius per year and rainfall is decreasing at a rate of 0.1 cm/year. They also estimate that, at current production levels , $\frac{\partial W}{\partial T} = -2$ and $\frac{\partial W}{\partial R} = 8$.

   a) What is the significance of the signs of these partial derivatives?

   b) Estimate the current rate of change of wheat production, $\frac{dW}{dt}$.

8. Use an implicit differentiation formula to find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$:

a) $x^2 + y^2 + z^2 = 3xyz$

b) $xyz = \cos(x + y + z)$